

Using Spot Instances for AI Training

Cutting costs with cloud's hidden discounts

A strategic approach to dramatically reduce your ML infrastructure costs without sacrificing performance.



Introduction: The Spot Instance Advantage



Unused Capacity

Spot instances represent unused cloud capacity sold at steep discounts compared to standard on-demand pricing



Massive Savings

70-90% cheaper than on-demand instances, transforming your cost structure for compute-intensive workloads



Training Focus

Ideal for training workloads that can handle occasional interruptions without compromising overall results

Our goal: **significantly reduce infrastructure spend** without sacrificing model performance or training capabilities

How Spot Instances Work

Spot instances operate on a fundamental premise: **the cloud provider can reclaim these resources at any time** when demand increases.

Preemption Reality

Your workload may be stopped with minimal notice (typically 30-120 seconds)

Provider Differences

AWS uses a market-based model, GCP offers fixed discounts, Azure uses priority-based allocation

Ideal Workloads

Works best with stateless, well-checkpointed training jobs that can resume progress





Benefits for AI Training

90%

Cost Reduction

Potential savings on GPU/TPU clusters compared to on-demand pricing

3x

Experiment Scale

Run more parallel experiments within the same budget constraints

24/7

Elastic Scaling

Dynamically expand compute during peak demand periods

↓ 60%

Budget Impact

Reduced barriers to entry for smaller teams needing high-end compute

Democratizing access to advanced compute resources for teams of all sizes

Challenges & Risks

Technical Considerations

■ Interruption Management

Training disruptions require graceful handling and efficient recovery protocols

■ Checkpoint Discipline

Without robust checkpointing, you risk losing hours of computation time

Business Limitations

■ Regional Variability

Spot availability varies significantly by region, instance type, and time

■ Inference Mismatch

Generally unsuitable for latency-sensitive inference workloads that need stability

 **Risk Assessment:** Always evaluate if your workload can tolerate interruptions before committing to a spot-based strategy. Some critical or time-sensitive training jobs may require on-demand guarantees.

Best Practices for Spot Success



Checkpoint Everything

Implement frequent model state saving in training loops (every N steps or epochs)



Hybrid Deployment

Mix on-demand instances (for critical components) with spot instances (for scalable training)



Use Management Tools

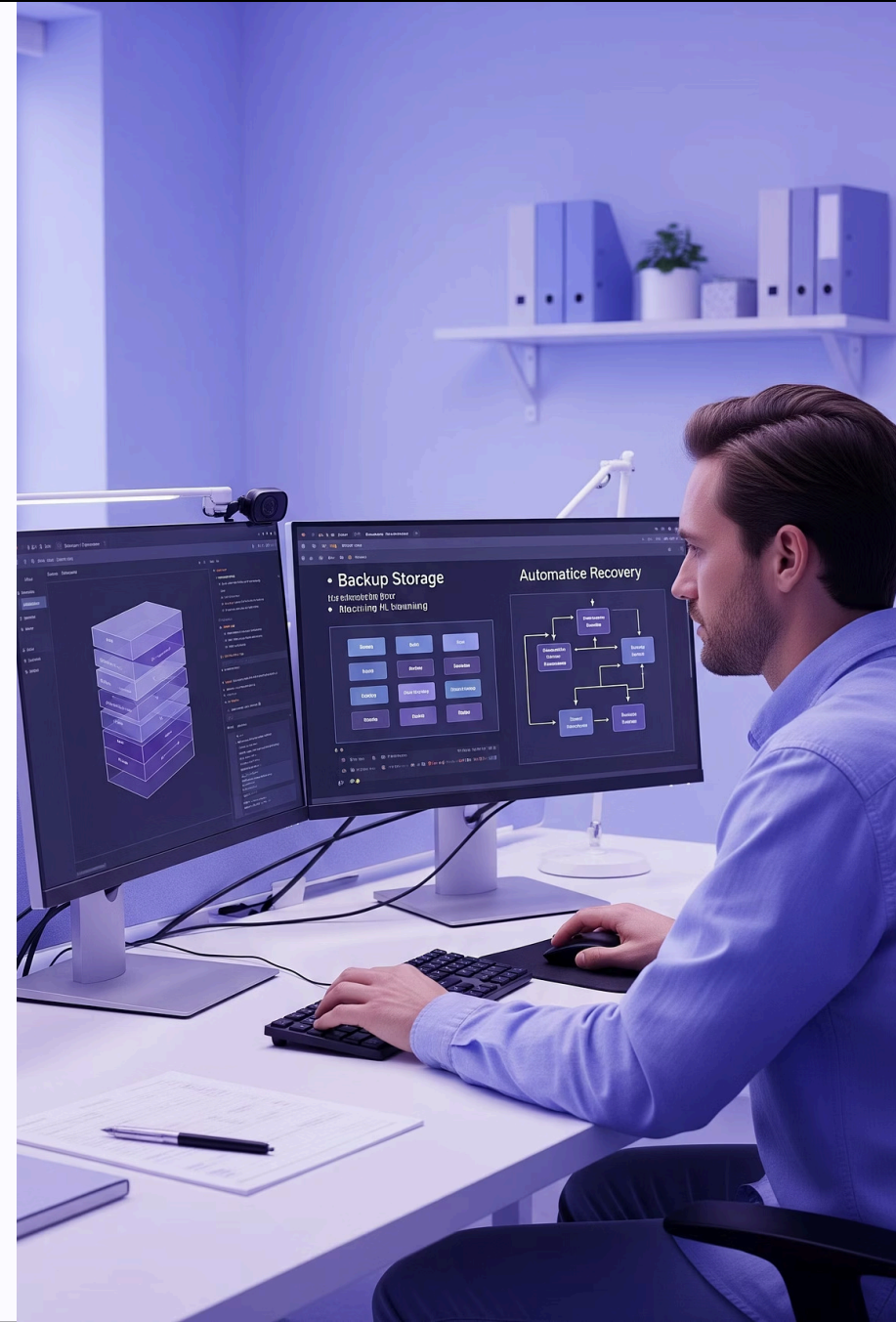
Leverage AWS Spot Fleet, GCP Preemptibles API, or Azure Spot VMs with orchestration



Monitor Availability

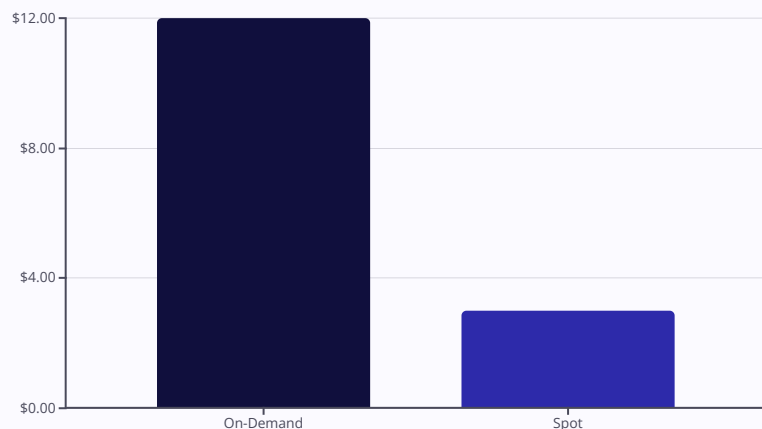
Track historical spot interruption rates and pricing before planning major workloads

A **well-architected spot strategy** requires both technical preparations and operational discipline



Real-World Example: ResNet Training on AWS

Cost Comparison: p3.2xlarge (V100 GPU)



75%

Total Savings

Reduction in total training costs
for multi-week jobs

4x

Experiment Volume

Increase in number of model
variants tested within budget

Performance Results

After implementing a spot-based training strategy with proper checkpointing:

- Achieved **identical model accuracy** as on-demand instances
- Total training time increased by only 15% despite interruptions
- Enabled running 3 parallel model variants instead of just 1
- Improved team's ability to experiment with hyperparameters

Key Innovation: Custom checkpoint manager that saves state every 100 batches with minimal overhead

Key Takeaways

Maximize Value, Minimize Waste

Smart Budgeting

Spot instances represent the most effective way to stretch ML infrastructure budgets, offering 70-90% savings with proper implementation

Workload Matching

Best suited for non-urgent, fault-tolerant training jobs where occasional interruptions won't compromise research goals

Technical Requirements

Demands serious investment in automation, checkpointing discipline, and recovery workflows to realize the full benefits

Strategic Importance

No longer optional—spot instances are now an essential component of cost-optimized AI infrastructure

Questions? Contact the ML Infrastructure team for implementation support

